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Jetson AI Lab



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Initial Setup using SDK Manager

Alternative setup method using NVIDIA SDK Manager to flash firmware and JetPack to your Jetson Orin Nano Developer Kit, including NVMe SSD installation support.

AUTHOR

Note

This guide is to supplement the official **Jetson Orin Nano Developer Kit Getting Started Guide**.

The NVIDIA® Jetson Orin Nano™ Developer Kit is a perfect kit to start your journey of local generative AI evaluation and development.

This guide explains the **alternative method** for setting up Jetson Orin Nano Developer Kit by flashing both the latest firmware (QSPI image) and the latest JetPack all at once, using a host x86 PC.

Check your toolbox

The following items are needed or highly desired for flashing your Jetson Orin Nano Developer Kit using your host PC.

If you don't have them, check the [default microSD-only setup instruction](#) or arrange them first then return to this guide once they are available.

⚠ What does NOT come in the box - What you need/want to prepare

Host PC

- ☐ x86 PC running **Ubuntu 22.04** or **Ubuntu 20.04**

Storage

Either of the following:

- ☐ microSD card (64GB or bigger)
- ☐ NVMe SSD (Recommended for better performance)



Flashing supply

- ☐ USB cable (USB-C plug for Jetson Orin Nano Developer Kit side, other end depends on your PC)
- ☐ Jumper pin (or metal paper clip)

Means to access terminal

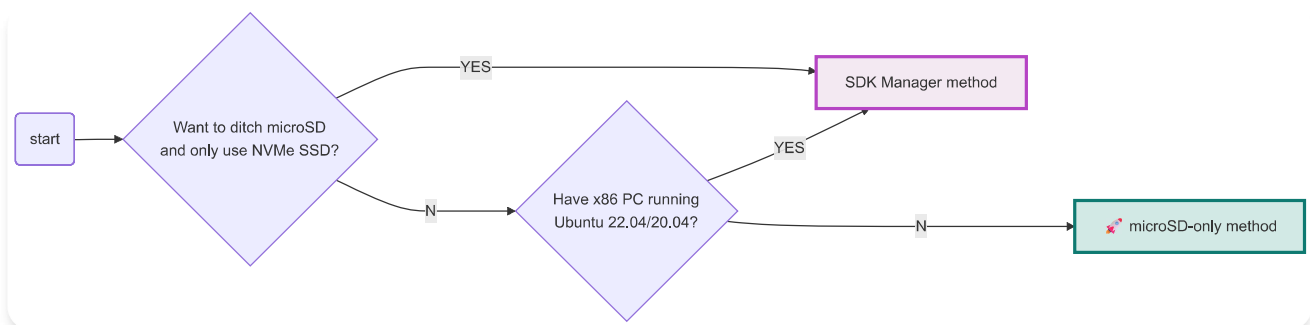
You need either of the following sets:

- ☐ DisplayPort cable, DisplayPort capable monitor and a USB keyboard
- ☐ DisplayPort to HDMI cable and HDMI capable monitor (or TV) and a USB keyboard
- ☐ **USB to TTL Serial cable** (Advanced)

Default method: microSD card only setup method

In case you **do NOT** have an x86 PC running Ubuntu 22.04 or 20.04, you can fall back to the default “microSD card only” setup method that does not require any host PC.

The decision process can look like this:



Click the button below to jump to the default setup page.



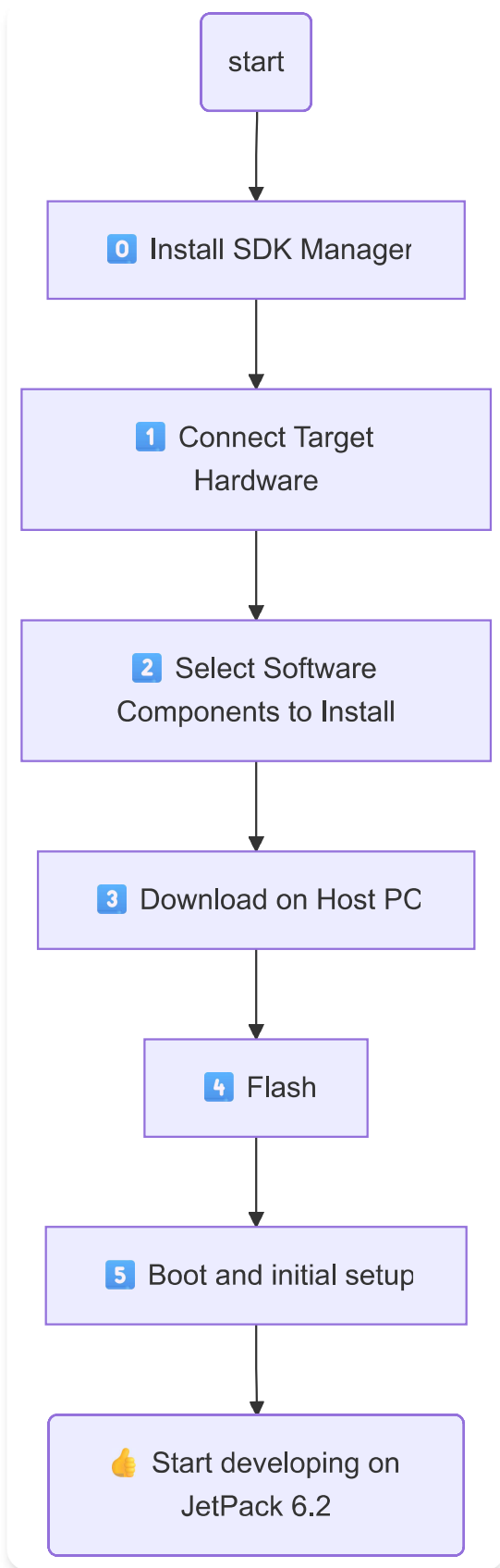
microSD-only method

Otherwise, continue reading on this page for the SDK Manager method.

Overall flow (SDK Manager method)



[Jetson Orin Nano Initial Setup Flowchart \(SDK Manager method\)](#)



0 Install SDK Manager

Reference

 [NVIDIA SDK Manager Documentation](#)

→ **Download and Install** → [Download and Run SDK Manager](#)

On your Ubuntu 22.04 / 20.04 PC, open a terminal and execute the following command.

Select the tab that matches your Ubuntu version.

 **Ubuntu 22.04**

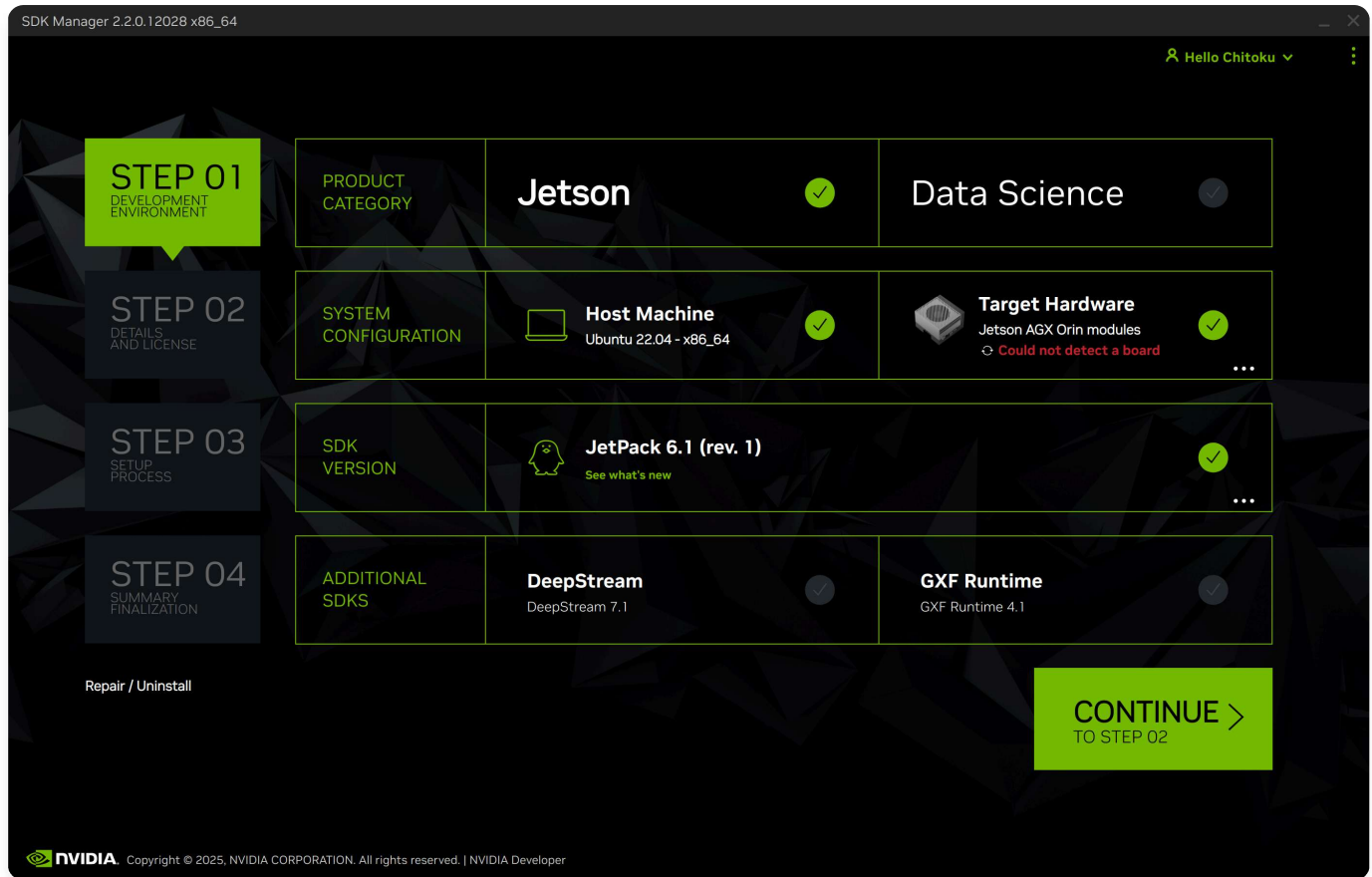
 Ubuntu 20.04

```
wget https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64/cuda-keyring_1.1-1_all.deb
sudo dpkg -i cuda-keyring_1.1-1_all.deb
sudo apt-get update
sudo apt-get -y install sdkmanager
```

Then launch SDK Manager with the following command:

```
sdkmanager
```

SDK Manager launched and the login completed.



Connect the Target Hardware

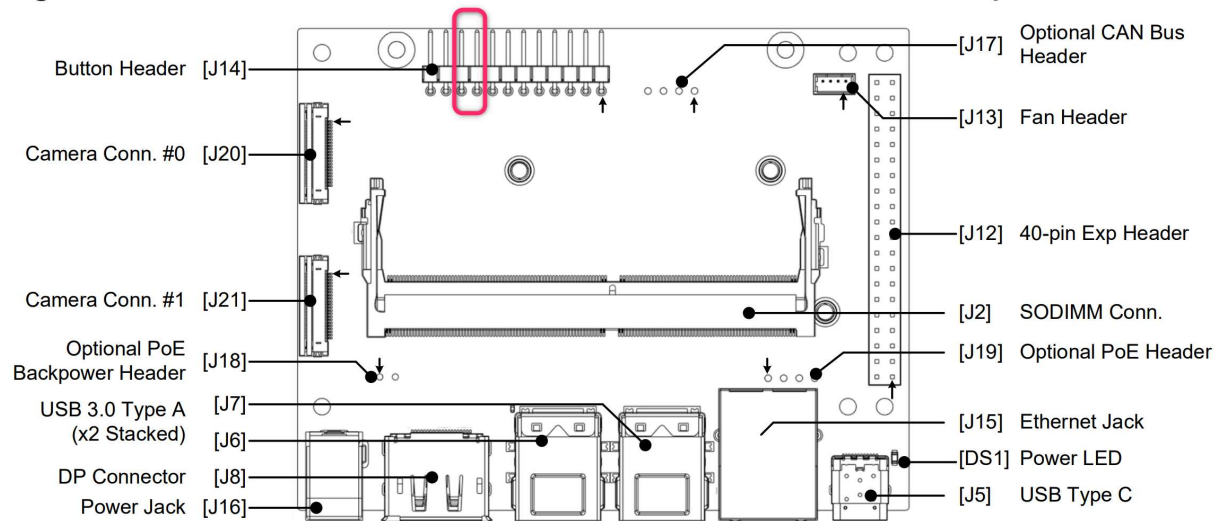
Once SDK Manager is launched, connect your Jetson developer kit to your Ubuntu PC and power it on in Forced Recovery mode.

1. **Connect** your Jetson Orin Nano Developer Kit to your PC with a USB cable.

"Note: The USB cable goes into the USB-C port of the carrier board of the developer kit."

2. While shorting pin 9 and pin 10 of J14 header located below the Jetson module using a jumper pin, insert the DC power supply plug into the DC jack of the carrier board to power it on.

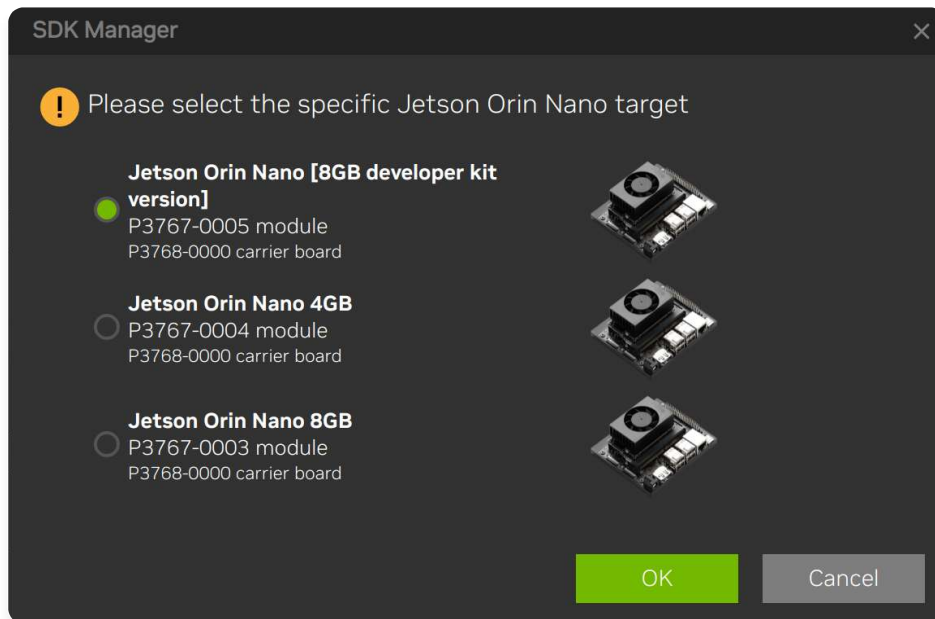
Figure 1-4. Jetson Orin Nano Carrier Board Placement – Top View



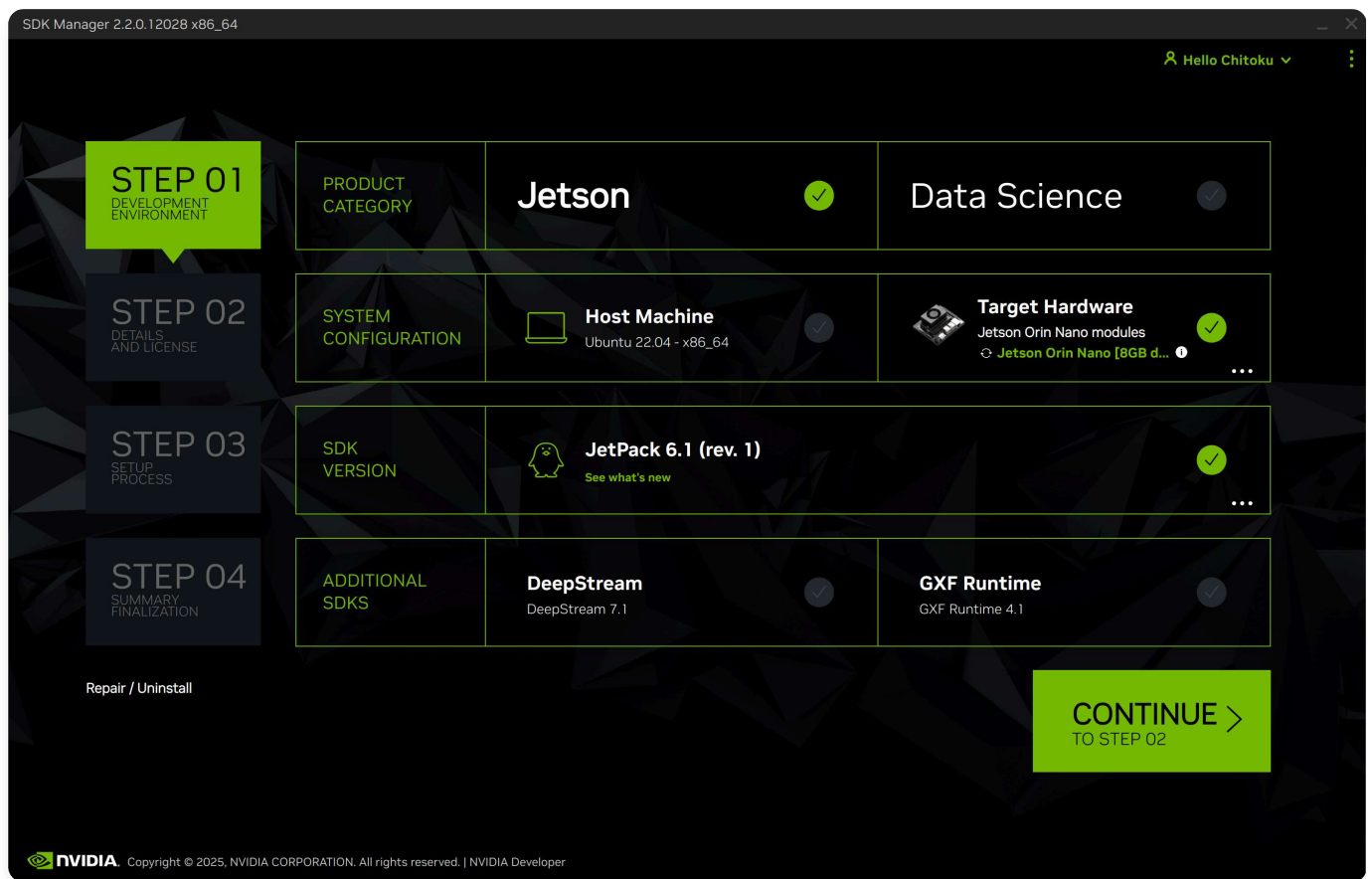
You can use a paper clip like the photo below as well.



3. Select "**Jetson Orin Nano [8GB developer kit version]**" and hit "**OK**"



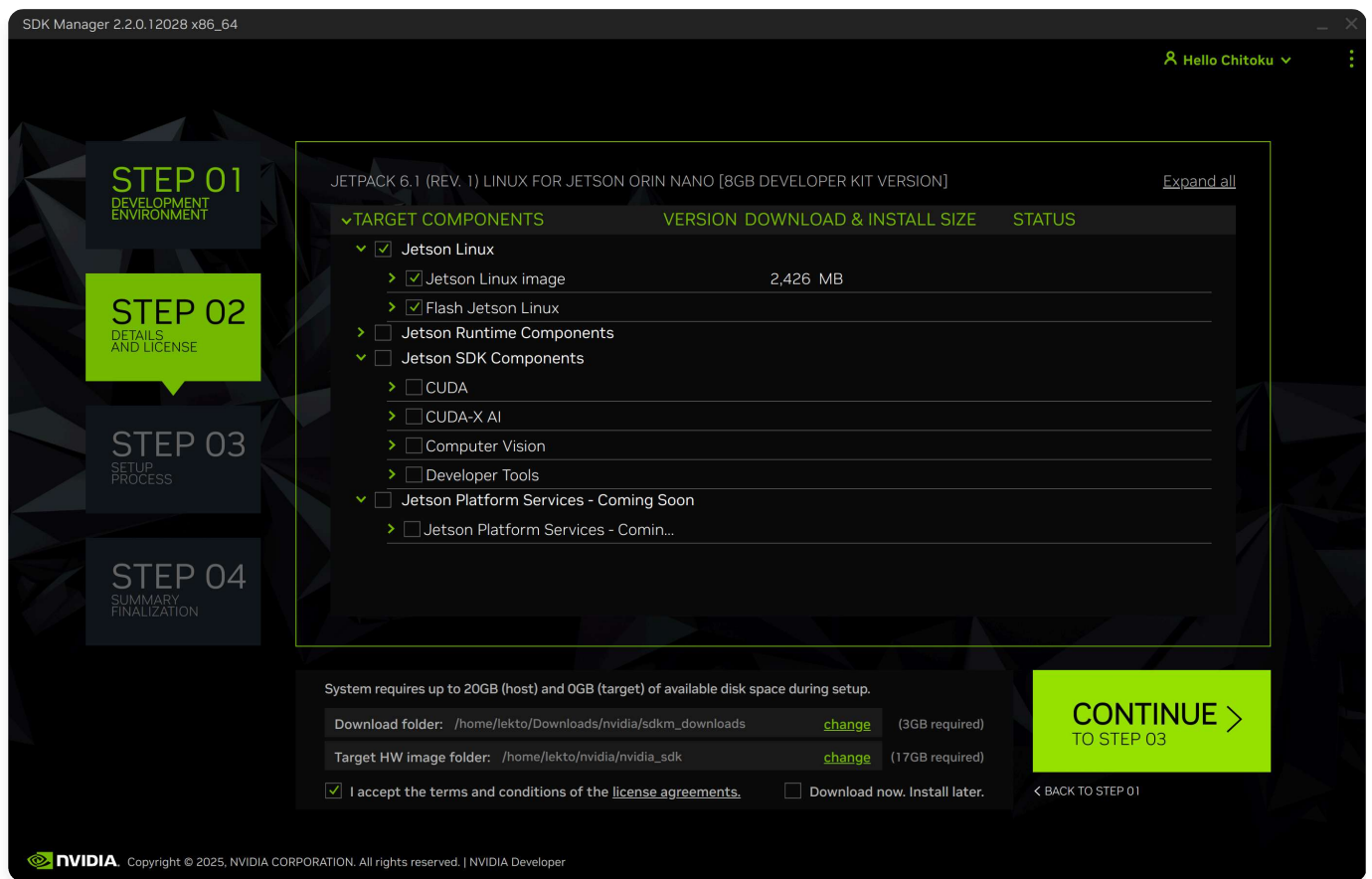
4. Uncheck "**Host Machine**" - it should now look like this:



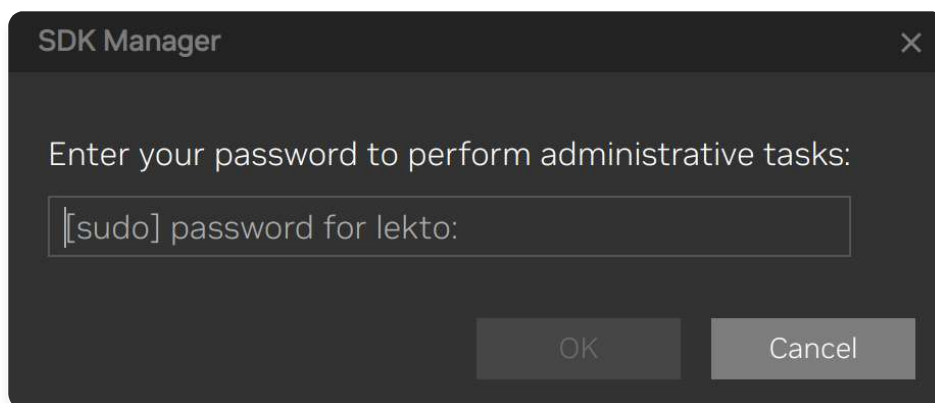
5. Click **"Continue"** button to proceed to the next step.

2 Select Software Components to Install

1. Leave only **"Jetson Linux"** component checked, and uncheck everything else.

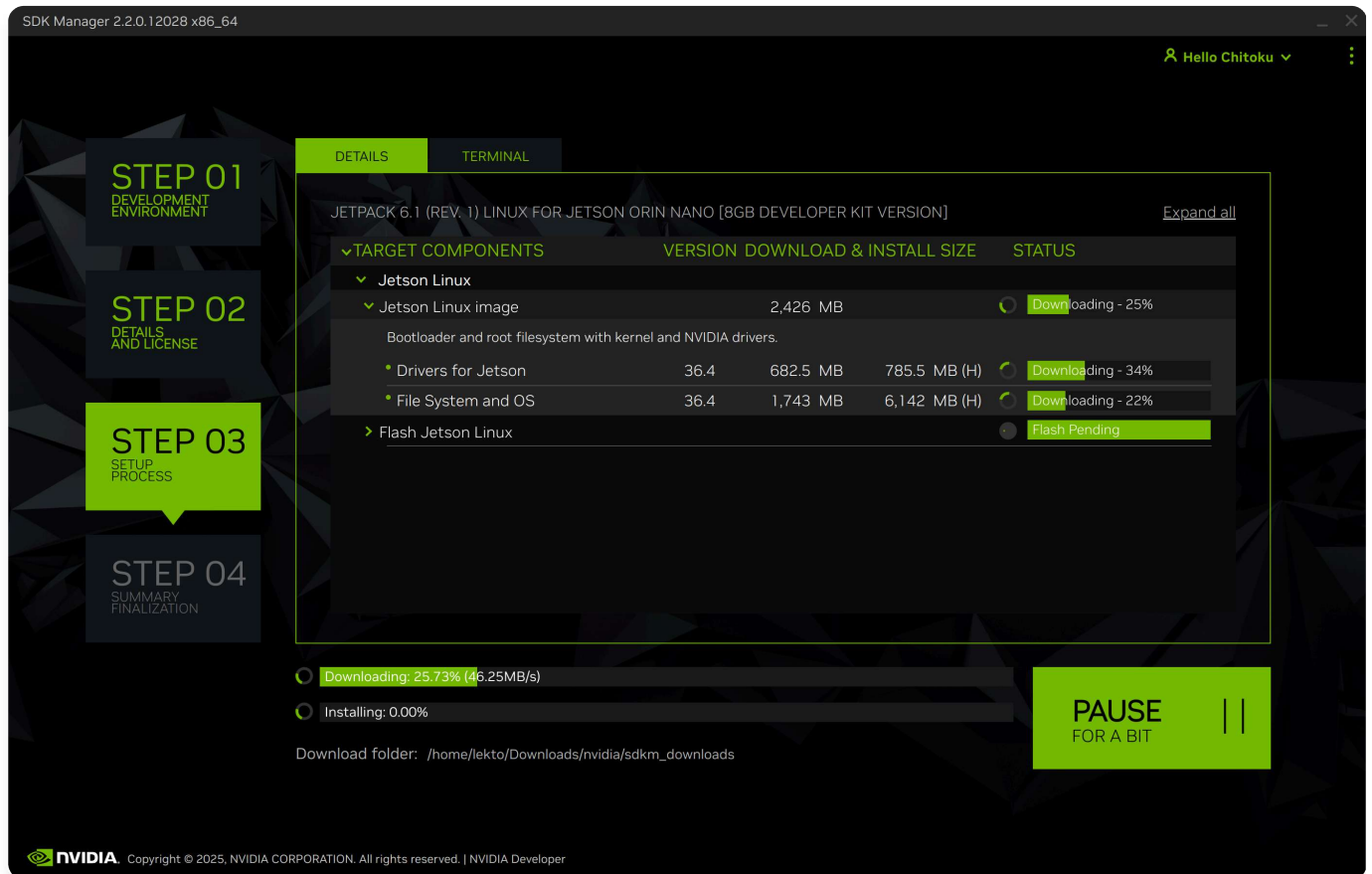


2. Remember to check **"I accept the terms and conditions of the license agreements"**.
3. Click **"Continue"** button to proceed.
4. It will prompt for the `sudo` command password.

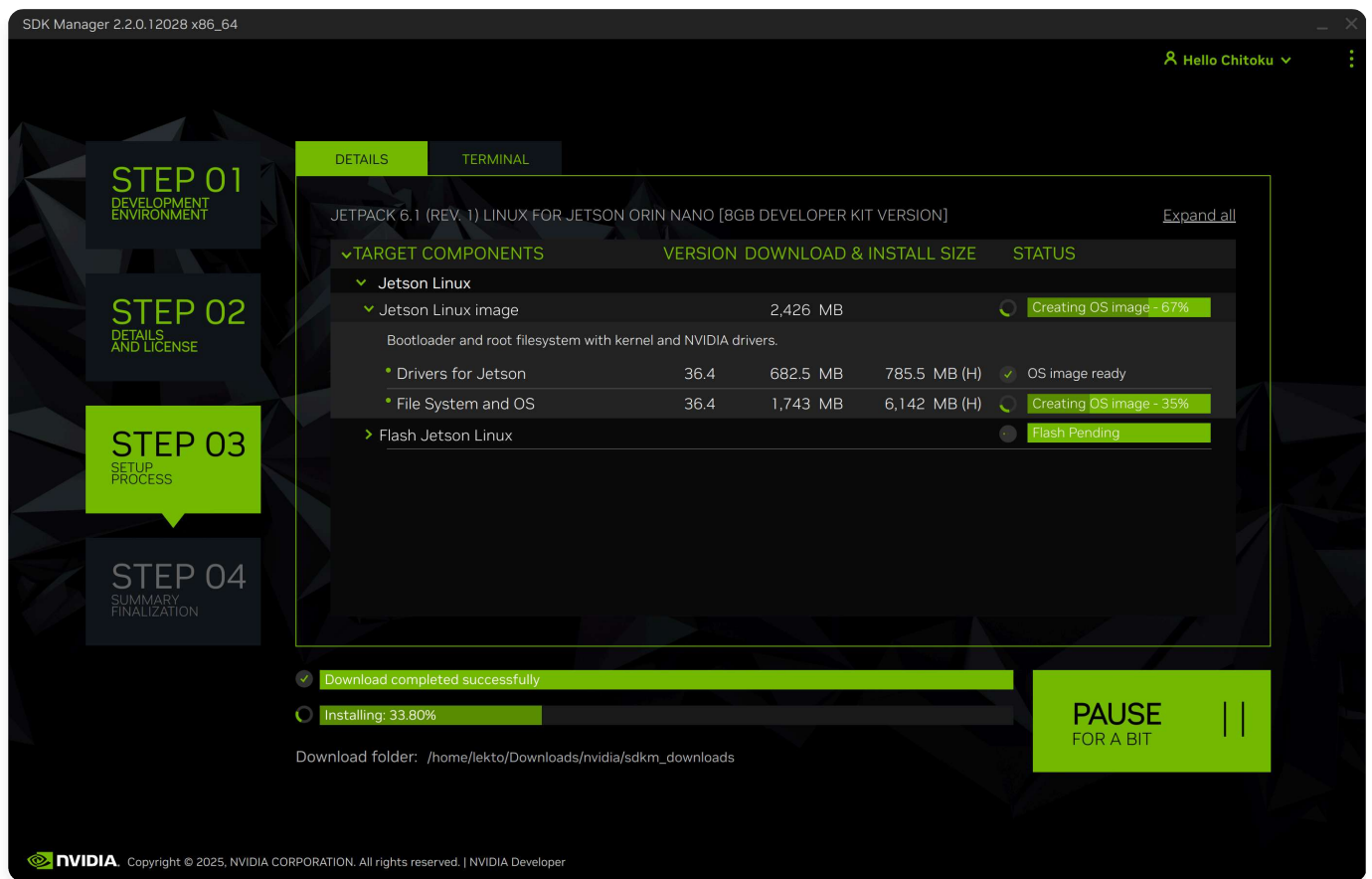


3 Download on Host PC

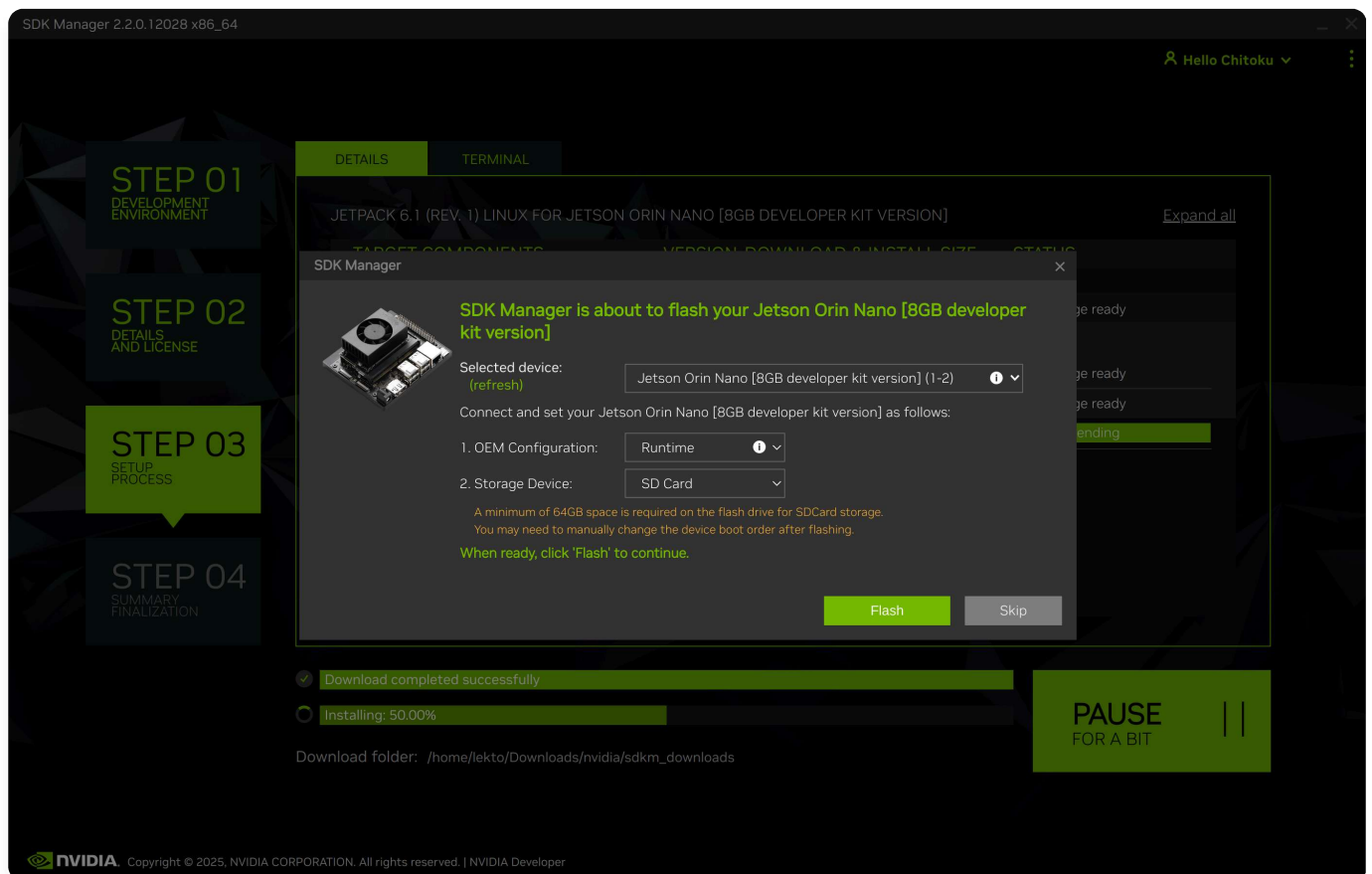
1. It will start downloading the "BSP" package and "RootFS" package.



2. Once downloads are complete, it will untar the package and start generating the images to flash in the background.

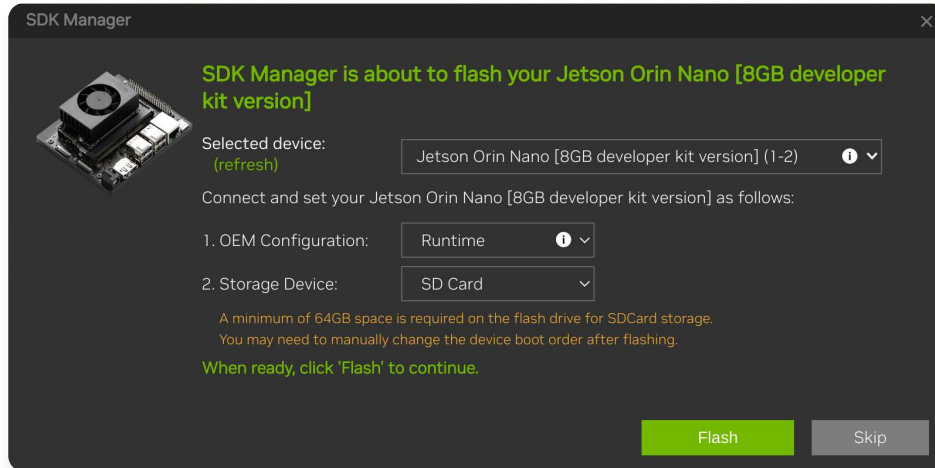


3. Once images are ready, SDK Manager will open the prompt for flashing.

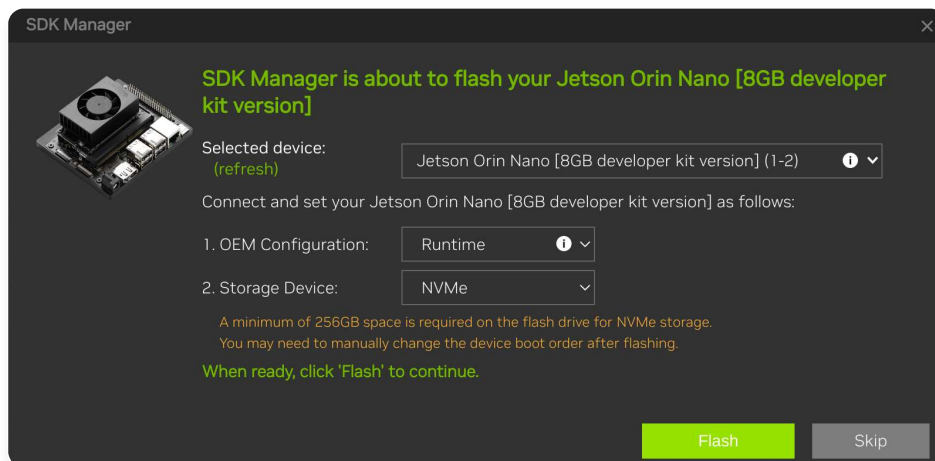


4 Flash

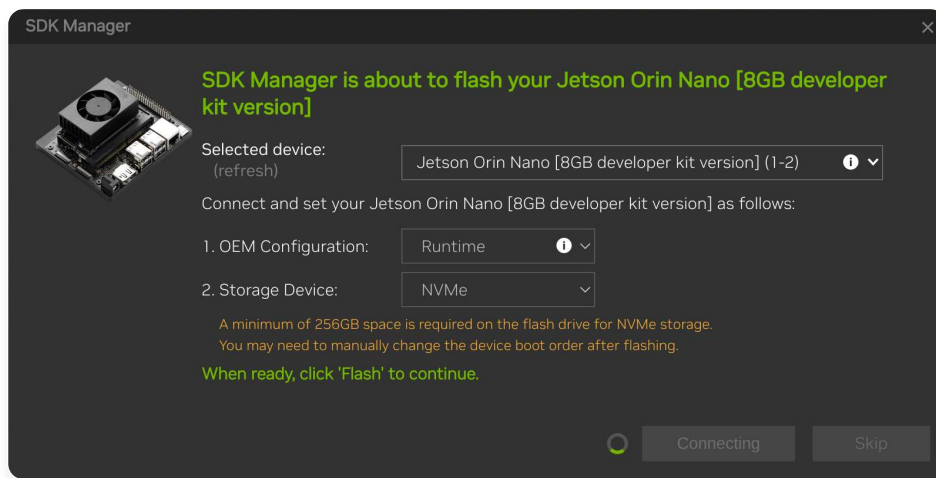
1. On the flashing prompt, select **"Runtime"** for "OEM Configuration".



2. Select **"NVMe"** if you want to flash Jetson Linux (BSP) to NVMe SSD (or **"SD Card"** for microSD).



3. Click **"Flash"** and the prompt popup will change.



4. Flash progress:

Details

Terminals

SDK Manager 2.2.0.12028 x86_64

Hello Chitoku

STEP 01
DEVELOPMENT
ENVIRONMENT

STEP 02
DETAILS
AND LICENSE

STEP 03
SETUP
PROCESS

STEP 04
SUMMARY
FINALIZATION

DETAILS

TERMINAL

JETPACK 6.1 (REV. 1) LINUX FOR JETSON ORIN NANO [8GB DEVELOPER KIT VERSION]

Expand all

TARGET COMPONENTS	VERSION	DOWNLOAD & INSTALL SIZE	STATUS
Jetson Linux			
Jetson Linux image		2,426 MB	OS image ready
Flash Jetson Linux			Flashing - 2%
Flash target device.			
Flash Jetson Linux	36.4	9,520 MB (H)	Flashing - 2%

Download completed successfully

Installing: 51.00%

Download folder: /home/leko/Downloads/nvidia/sdkm_downloads

PAUSE
FOR A BIT

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5. Flash successfully completes:

SDK Manager 2.2.0.12028 x86_64

Hello Chitoku

STEP 01
DEVELOPMENT ENVIRONMENT

STEP 02
DETAILS AND LICENSE

STEP 03
SETUP PROCESS

STEP 04
SUMMARY FINALIZATION

DETAILS | **TERMINAL**

JETPACK 6.1 (REV. 1) LINUX FOR JETSON ORIN NANO [8GB DEVELOPER KIT VERSION] [Expand all](#)

TARGET COMPONENTS	VERSION	DOWNLOAD & INSTALL SIZE	STATUS
✓ Jetson Linux			
> Jetson Linux image		2,426 MB	✓ OS image ready
✓ Flash Jetson Linux			✓ Success
Flash target device.			
• Flash Jetson Linux	36.4	9,520 MB (H)	✓ Success

✓ **INSTALLATION COMPLETED SUCCESSFULLY.** [EXPORT LOGS](#)

FINISH AND EXIT ✕

< BACK TO STEP 01

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5 Boot and Initial Setup

1. If still plugged, **remove the jumper** from header (that was used to put it in Forced Recovery mode)
2. **Connect peripherals:** DisplayPort cable/adaptor and USB keyboard/mouse to Jetson Orin Nano Developer Kit, or hook up the USB to TTL Serial cable.
3. **Power cycle:** Unplug the power supply and put back in.

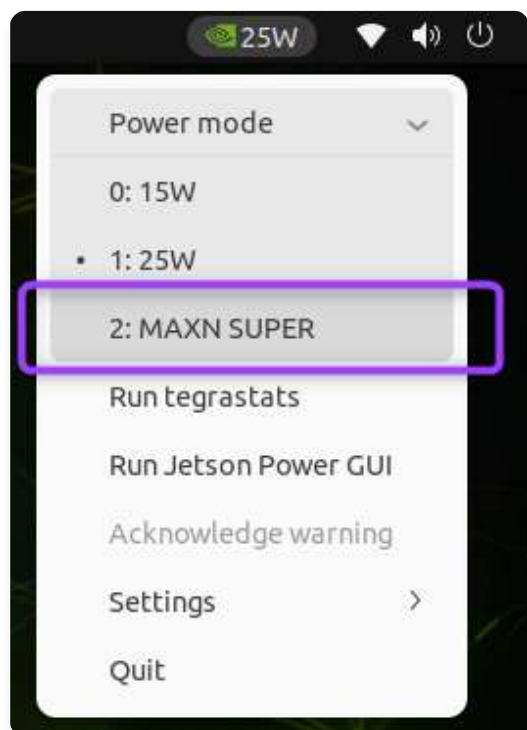
4. Jetson should now boot into the Jetson Linux (BSP) of your selected JetPack version from the storage of your choice.
 5. **Complete the initial software setup** (`oem-config`)
-

Switch to MAXN SUPER mode

Note that the default power mode on JetPack 6.2 on Jetson Orin Nano Developer Kit is **25W**.

To switch to the **MAXN SUPER** mode and unlock the unregulated performance:

1. Click on the current power mode (**25W**) by clicking the NVIDIA icon on the right side of the Ubuntu desktop's top bar.
2. Select **Power mode** from the menu.
3. Choose **MAXN SUPER** to enable maximum performance.





Congratulations!

Your Jetson Orin Nano Developer Kit is set up with JetPack 6.2 image and you are ready to develop!

Next Steps

- [SSD + Docker Setup](#) - Install NVMe SSD and configure Docker
- [microSD-only method](#) - Alternative setup without host PC

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